CS4001- Professional Practices in IT

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Management Accounting

After reading this chapter, you should, in simple cases:

- *be able to calculate the cost of labor;*
- *understand the concept of overheads and the different ways in which they may be distributed;*
- *be able to determine how much it costs to produce a particular product or provide a specific service;*
- *understand how to produce a budget and a cash flow forecast, and how to monitor them*

COST OF LABOUR

- The cost of employing someone is more than just the cost of their salary.
- In most countries, employers are required to pay a tax for every employee. This tax usually goes by a name such as *employers' national insurance contribution* (in the UK) or *social security contribution; it is proportional to the* employee's salary.
- In some countries, this contribution may be as large as 60 per cent of salary, while in others it is very much smaller.

Scenario - Example

- Suppose you decide to set up a company that makes computers. The company buys items like processor boards, memory chips, hard disks, etc.
- It then assembles the machines, installs and configures the software, and delivers the computer to the customer. There is, in principle, no problem in working out the cost of the components that go into the computers –although it is only too easy to forget to add in the cost of the odd small component.
- What is more difficult is working out the cost of labor

Cost of labor

TABLE 7.1 *Calculation of the number of revenue-earning hours in a year*	
Total number of weekdays (1)	260
Public holidays (2)	10
Annual leave (3)	20
Sick leave (4)	5
Unproductive time (5)	10
Total non-revenue-earning time (2) + (3) + (4) + (5) = (6)	45
Total number of revenue earning days (1) – (5) = (7)	215
Total number of hours available (7) × 7	1505

Pay roll

The total cost of employing a person, that is, the salary plus employers' social security contributions plus any other costs associated directly with the employee, is sometimes known as the employee's *payroll cost or direct cost*.

Example :

- *Consider a technician who is employed* to assemble the computers and suppose that he is paid an annual salary of £20,000. His payroll costs are £22,000.
- The direct cost of the technician is £22,000 per year so the direct cost of an hour of their time is £22000/1505 = £14.62.

OVERHEADS

- Costs that cannot be directly associated with a particular product are known as overheads.
- There are many costs that we have ignored. Even if our technician works as a sole trader, with no assistance, there will be other costs he has to pay for a computer costs £200 and it takes 10 hours of a technician’s time to assemble the computer, install the software and configure it, then the cost of constructing the finished product is £200 + 10 × £14.62 = £346.20.

Overheads

TABLE 7.2 *Direct costs and expected sales for the different computer models*

Model	Cost of components (£)	Technician time (hours	Expected sales
Basic	£200	10 hours	200
Advanced	£300	12 hours	100
Professional	£400	15 hours	50

Allocate the overhead to each computer sold

- Expect to sell 350 units, this means £181.43 per computer(overhead calculated from budget total over head is 63500 in table 7.4). This
- Basic model would cost:

  £181.43 + £200 + 10 × £14.62 = £527.63
- Advanced model would cost:

  £181.43 + £300 + 12 × £14.62 = £651.83
- Professional model would cost:

  £181.43 + £400 + 15 × £14.62 = £802.23

The second way of allocating the overhead

- The second way of allocating the overhead is to make it proportional to the number of hours of labor involved. This means adding an overhead component to the cost of an hour of a technician's time. Since we have three technicians, each supplying 1,505 hours of productive labor per year, we need to add

  £63500/(3 × 1505) = £14.06

- Cost of an hour's labor = £14.62 + £14.06 = £28.68.
- Now, the cost of the three models comes out at:
- £200 + 10 × £28.68 = £486.80 (Basic)
- £300 + 12 × £28.68 = £644.16 (Advanced)
- £400 + 15 × £28.68 = £830.20 (Professional)

TABLE 7.3 *Effects of different overhead calculations*

	Fixed overhead	Overhead proportional to labour content	Overhead proportional to total direct cost
Basic	528	487	495
Advanced	652	644	680
Professional	802	830	886

BUDGETING

- A budget is a financial plan showing the expected income and expenditure for an organization over a specific period, typically one year.

Example

TABLE 7.4 *An example budget*

Overhead Expenditure	
Owner's payroll costs	42,000
Secretary's payroll costs (part time)	8,000
Costs of van (including depreciation)	3,500
Internet connection, telephone, postage, etc.	1,000
Advertising	2,000
Premises (heating and lighting, rent, rates, etc.)	4,500
Professional fees	1,000
Insurance	500
Total overheads	63,500
Operating costs	
Technicians' payroll costs	66,000
Bought-in components	90,000
Total manufacturing costs	156,000
Total costs	219,500
Sales income	
Basic model (200 @ £595)	119,000
Advanced model (100 @ £795)	79,500
Professional model (50 @ £895)	44,750
Total sales	243,250
Profit	24,750

BUDGETING

- For many reasons, particularly with taxation and social security, owners should treat themselves as employees and pay themselves a salary, rather than attempt to live on the company's profits. This accounts for the item labeled 'Owner's payroll costs'.
- The services of an accountant will be necessary to help prepare the annual accounts and possibly to advise from time to time. The advice lawyer may also be necessary from time to time. These items are covered under the heading 'Professional fees'.
- Finally, employers are legally required to carry insurance to cover any claim against them for injuries suffered by employees during their employment; other insurance, against theft from the company's premises for example, may also be necessary. This explains the heading 'Insurance'.

CASH FLOW FORECAST

A company may be very profitable but unable to pay its bills. For that reason, it may be forced into receivership. This apparent paradox typically arises because bills have to be met, in particular staff have to be paid, before the income they generate is received. To avoid this difficulty businesses, need to prepare cash flow forecasts, that is, estimates of the amount of money that will flow into and out of the company each month.

companies normally try to forecast twelve months.